PROJECT AND TEAM INFORMATION

Project Title

PulsePath : Smart Routing for Faster Emergency Ambulance Response

Student / Team Information

Team Name:	***Code Prodigies***
Team member 1 (Team Lead)	*Negi,Karan – 2510010316* *karannegi8494@gmail.com* 
Team member 2	*Singh,Tanveer – 2510012529* *tanveersaini@gmail.com* 
Team member 3	*Singh Rawat,Ronak – 2510018478* *ronakrawat512@gmail.com* 
Team member 4	*Joshi,Yukta – 2510010020* *joshiyukta18@gmail.com* 

PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

(Describe the problem you want to solve and why it is important. Max 300 words)

As we know emergencies can happen to anyone anytime. therefore we need to be already prepared for it. in such situations the ambulance need to go to the location as fast as possible. But while doing so, there's many problems we face. such as: traffic jams, road blocks,ambulance not free,etc..
we chose this project so we can develop a system that provides a suitable and available ambulance and provide the fasted route to the emergency location.
we will make a city road network which would be represented as graph. where different locations would be nodes and roads would be weighted edges. the weight of the edge will be the estimated time needed to reach that particular location.
the system will also check the available ambulances and find out which one would reach the location fastest.and after reaching the patient it will also find the nearest hospital and the shortest route there.
one more important part of this project is that it will adapt to road changes also. if the traffic increases or a road closes, then the road information would be updated in the system. and it will again calculate the route.
through this project we will understand graphs,data structures and shortest path algorithms in a practical way.

State of the Art / Current solution (1 pt)

(Describe how the problem is solved today (if it is). Max 200 words).

Ambulance services use GPS,communication centers and routing systems to manage and guide the ambulance to the emergency location.
In simple systems the ambulance can be assigned manually or a nearest available ambulance can be selected. But the nearest ambulance doesn't always mean the fastest travel.
Due to traffic,accidents and road closures the travel time can be increased. and the selected route can become less suitable. The routing system can find the shortest route by representing road network as graph and using shortest path algorithms.
our project also follows this graph based approach, but its focus is on ambulance selection and emergency routing. instead of just considering the distance it considers availability of ambulance and travel time. it uses Dijkstra's algorithm to find the shortest route.
if the road conditions change. then the road weight can be updated and the path will be recalculated.

Project Goals and Milestones (2 pts)

(Describe the project general goals. Include initial milestones as well any other milestones. Max 300 words).

The primary goal of this project is to develop a smart ambulance dispatch and routing system that helps reduce the time required to respond to an emergency. The system will model the city road network as a weighted graph and use efficient graph algorithms to make routing decisions. The main goals of the project are:

- *To identify and select the optimal available ambulance for an emergency.*
- *To calculate the minimum travel-time route using Dijkstra's algorithm.*
- *To find the nearest hospital after the ambulance reaches the patient.*
- *To handle changes such as traffic congestion and road closures through dynamic route updates.*
- *To use efficient data structures such as adjacency lists and priority queues for better performance.*
- *To reconstruct and display the actual path between locations.*

The project will be completed through several milestones. The first stage involves designing the city road network using graph representation. Next, ambulance and hospital information will be integrated into the system. The shortest-path engine based on Dijkstra's algorithm will then be implemented. After this, ambulance selection and hospital selection modules will be developed. A dynamic traffic handling feature will be added to update road conditions. Finally, the complete system will be tested using different emergency scenarios, including unavailable ambulances and closed roads.

Project Approach (3 pts)

(Describe how you plan to articulate and design a solution. Including platforms and technologies that you will use. Max 300 words).

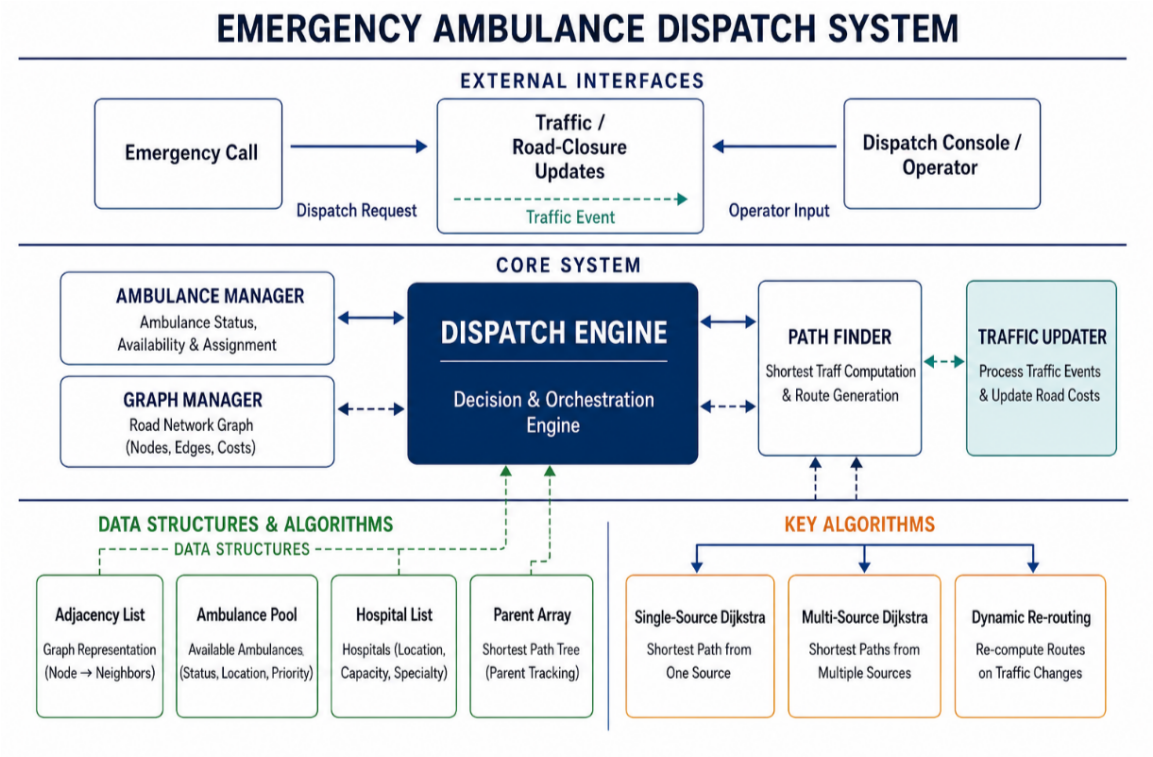
The system will be developed in C++ using graphs and shortest-path algorithms. The city road network will be stored as a weighted graph. Each location will be represented by a node and each road by an edge. The weight of each edge will represent the estimated travel time.

We will use an adjacency list to store the graph because it is suitable for a road network where most locations are connected to only some of the other locations. A priority queue will also be used with Dijkstra's algorithm to select the location with the lowest current travel time. When an emergency request is received, the system will first check which ambulances are free then calculate the travel time from each available ambulance to the emergency location and select the ambulance with the minimum travel time.

Once the ambulance reaches the patient, the system will then calculate routes to the available hospitals and select the one with the minimum travel time. A parent array will be used to reconstruct the actual route. If traffic changes or a road is closed, the corresponding road weight will be updated. The system will then run the shortest-path calculation again to find a better route.

System Architecture (High Level Diagram)(2 pts)

(Provide an overview of the system, identifying its main components and interfaces in the form of a diagram using a tool of your choice).



Project Outcome / Deliverables (1 pts)

(Describe what are the outcomes / deliverables of the project. Max 200 words).

The project is expected to provide a practical solution for improving ambulance dispatch and emergency routing using graph algorithms. The system will focus on selecting an available ambulance, finding the fastest route, and adapting to changing road conditions. The main deliverables of the project are:

- *A functional C++ based prototype for smart ambulance dispatch and routing.*
- *A weighted graph representation of the city road network.*
- *A system to track ambulance locations and availability status.*
- *Automatic selection of the most suitable free ambulance.*
- *Fastest route calculation using Dijkstra's shortest-path algorithm.*
- *Identification of the nearest hospital based on travel time.*
- *Dynamic route recalculation for traffic changes and road closures.*
- *Efficient implementation using adjacency lists and priority queues.*

Assumptions

(Describe the assumptions (if any) you are making to solve the problem. Max 100 words)

The system is developed based on the following assumptions:

- *All road travel times are non-negative.*
- *Ambulance locations and availability status are updated correctly.*
- *Hospital locations are predefined in the system.*
- *The city road network is accurately represented as a weighted graph.*
- *Traffic changes and road closures are provided as updates to road weights.*
- *The current system handles one emergency request at a time.*
- *A valid route exists between the required locations.*

References

(Provide a list of resources or references you utilised for the completion of this deliverable. You may provide links).

1. *E. W. Dijkstra, "A Note on Two Problems in Connexion with Graphs," Numerische Mathematik, vol. 1, pp. 269–271, 1959.*

2. *T. H. Cormen, C. E. Leiserson, R. L. Rivest, and C. Stein, Introduction to Algorithms, MIT Press.*

3. *B. Stroustrup, The C++ Programming Language, Addison-Wesley.*

4. *GeeksforGeeks – Dijkstra's Shortest Path Algorithm using Priority Queue. available on : https://www.geeksforgeeks.org/dsa/dijkstras-shortest-path-algorithm-using-priority$_q$ueue − stl/*

5. *OpenStreetMap.Available : https://www.openstreetmap.org*